Simulation Example for Spinach

This is a basic example to run AquaCrop from `R` for the spinach crop.

Importantly, the working directory should be set to the folder in which the *aquacrop.exe* file is located. This path should also contain **no spaces**. For example: `path_to_aquacrop_folder <- "C:/users/username/AquaCrop71/"`.

```
library(AquacropOnR)
library(tidyverse)
setwd(dir = path_to_aquacrop_folder)
```

With the package comes an example list with default crop parameters for **quinoa** and **spinach**. To make this list for your own crop, you need an AquaCrop cropfile (*YourCrop.CRO*) whose path should be input in the `read_CRO()` function:

```
yourcrop <- read_CRO("path/YourCrop.CRO")`
```

Once you have the list with default crop parameter values, you can design the scenario's for which you want to run AquaCrop. The `AquacropOnR` package provides a function to design the scenario's: `design_scenario()`. The arguments in this function have the following meaning:

- `name` is a character vector of names for the scenario's.
- `Input_Date` is a Date vector that indicates the start point of the meteo files (`Plu`, `Tnx`, `ETo`).
- `Sim_Date` is a Date vector that indicates the start point of the simulation in each scenario.
- `Plant_Date` is a Date vector that defines the planting date in each scenario.

- `IRRI` is a character vector of the names of the irrigation scenario's present in the `ID` column of the `IRRI_s` tibble.

- `Soil` is a character vector of the names of the soils present in the `ID` column of the `SOL_s` tibble.

- `Plu`, `Tnx`, and `ETo` are character (vectors) that refer to the names of `R` objects, that hold the daily data for precipitation, temperature and reference evapotranspiration, respectively.
- `FMAN` is a character vector of the names of the management scenario's present in the `ID` column of the `FMAN_s` tibble.
- `SWO` is a character vector of the names of the initial condition scenario's present in the ID column of the `SWO_s` tibble.
- `GWT` is a numeric vector of values for the depth of the Ground Water Table (in m), for each scenario.

IMPORTANT:

- The Scenario_s tibble **must** be named `Scenario_s` and **must** have the columns (variables) `Scenario`, `Input_Date`, `Plant_Date`, `IRRI`, `Soil` and `Meteo`. Other arguments `Sim_Date`, `FMAN`, `GWT` and `SWO` have default values if not provided explicitly. See `design_scenario()`.

- The irrigation tibble **must** be named `IRRI_s` and **must** have the columns `ID`, `Timing`, `Depth` and `ECw`. Values of the `ID` column are given as input to the `IRRI` argument in the `design_scenario()` function.

- The soil tibble **must** be named `SOL_s` and **must** have the columns `ID`, `Horizon`, `Thickness`, `SAT`, `FC`, `WP`, `Ksat`, `Penetrability` and `Gravel`. Values of the `ID` column are given as input to the `Soil` argument in the `design_scenario()` function.

- The precipitation tibble can be named as you want, but its name is the input to the `Plu` argument in the `design_scenario()` function. This tibble **must** have the columns `DAY` and `PLU`.

- The reference evapotranspiration tibble can be named as you want, but its name is the input to the `ETo` argument in the `design_scenario()` function. This tibble **must** have the columns `DAY` and `ETo`.

- The temperature tibble can be named as you want, but its name is the input to the `Tnx` argument in the `design_scenario()` function. This tibble **must** have the columns `DAY`, `TMAX` and `TMIN`.
- The Management tibble **must** be named `FMAN_s` and **must** have the columns `ID`, `mulch_perc`, `mulch_eff`, `fert_stress`, `soil_bunds`, `runoff_aff`, `CN_eff`, `weed_clo`, `weed_mid`, `weed_cc`.

- The initial conditions tibble **must** be named `SWO_s`, and **must** contain the columns `ID`, `Horizon`, `Thickness`, `WC` and `ECe`.

Examples of these tibbles are available from the package.

The default `FMAN_s` tibble associated to the package looks like this:

```
#> # A tibble: 1 x 10
#>   ID      mulch_perc mulch_eff fert_stress soil_bunds runoff_aff CN_eff weed_clo
#>   <chr>        <dbl>     <dbl>       <dbl>      <dbl>      <dbl>  <dbl>    <dbl>
#> 1 default          0        50           0          0          0      0        0
#> # i 2 more variables: weed_mid <dbl>, weed_cc <dbl>
```

but you can create your own, as long as it has the same column names, and is named `FMAN_s`. The default `IRRI_s` tibble associated to the package looks like this:

```
#> # A tibble: 2 x 4
#>   ID       Timing Depth   ECw
#>   <chr>     <dbl> <dbl> <dbl>
#> 1 no_irrig     20     0     0
#> 2 simple       20    25     0
```

We can make a new irrigation input `tibble` as follows:

```
IRRI_s <- IRRI_s %>%
  bind_rows(tibble(ID = c("IRRI_01", "IRRI_01"), Timing = c(20, 35),
                   Depth = c(25,25), ECw = c( 0, 0)))
```

which looks like:

```
#> # A tibble: 4 x 4
#>   ID       Timing Depth   ECw
#>   <chr>     <dbl> <dbl> <dbl>
#> 1 no_irrig     20     0     0
#> 2 simple       20    25     0
#> 3 IRRI_01      20    25     0
#> 4 IRRI_01      35    25     0
```

For this example we will use the data for soil, precipitation, temperature and reference evapotranspiration that comes with the package. These tibbles should be formatted as presented below.

```
#> # A tibble: 1 x 9
#>   ID      Horizon Thickness   SAT    FC    WP  Ksat Penetrability Gravel
#>   <chr>     <dbl>     <dbl> <dbl> <dbl> <dbl> <dbl>         <dbl>  <dbl>
#> 1 Soil_01       1         4    41    22    10  1200           100      0
#> # A tibble: 6 x 2
#>     DAY   PLU
#>   <dbl> <dbl>
#> 1     1   0.4
#> 2     2   0.7
#> 3     3   0.8
#> 4     4   0.2
#> 5     5   0  
#> 6     6   0.3
#> # A tibble: 6 x 3
#>     DAY  TMAX  TMIN
#>   <dbl> <dbl> <dbl>
#> 1     1   9.3   7.2
#> 2     2   7.4   5  
#> 3     3   6.8   2.1
#> 4     4   6.5   2.5
#> 5     5   6.5   2.9
#> 6     6   7.4   4.3
#> # A tibble: 6 x 2
#>     DAY   ETo
#>   <dbl> <dbl>
#> 1     1 1.03 
#> 2     2 1.17 
#> 3     3 0.743
#> 4     4 0.887
#> 5     5 0.778
#> 6     6 0.695
```

```
Scenario_s <- design_scenario(
  name = c("S_01", "S_02"),
  Input_Date = as.Date("2019-01-01"),
  Plant_Date = as.Date(c("2019-04-01", "2019-04-01")),
  IRRI = c("no_irrig", "IRRI_01"),
  Soil = "Soil_01",
  Plu = "Plu_01",
  Tnx = "Tnx_01",
  ETo = "ETo_01",
  FMAN = "default",
```

```r
  CO2 = "CO2_01",
  GWT = 2.0
  )
```

The `Scenario_s` tibble should look like this:

```
#> # A tibble: 2 x 13
#>   Scenario Input_Date Sim_Date   Plant_Date IRRI   Soil  Plu   Tnx   ETo   CO2
#>   <chr>    <date>     <date>     <date>     <chr>  <chr> <chr> <chr> <chr> <chr>
#> 1 S_01     2019-01-01 2019-04-01 2019-04-01 no_ir~ Soil~ Plu_~ Tnx_~ ETo_~ CO2_~
#> 2 S_02     2019-01-01 2019-04-01 2019-04-01 IRRI_~ Soil~ Plu_~ Tnx_~ ETo_~ CO2_~
#> # i 3 more variables: FMAN <chr>, GWT <dbl>, SW0 <chr>
```

Then it is crucial to create the correct AquaCrop path, while checking the required folders and choosing which daily outputs to produce. Therefore the package has the function `path_config()`. Make sure your path ends with a "/". For example: `path_to_aquacrop_folder <- "C:/users/username/AquaCrop71/"`.

```r
AQ <- path_config(AquaCrop.path = path_to_aquacrop_folder)
```

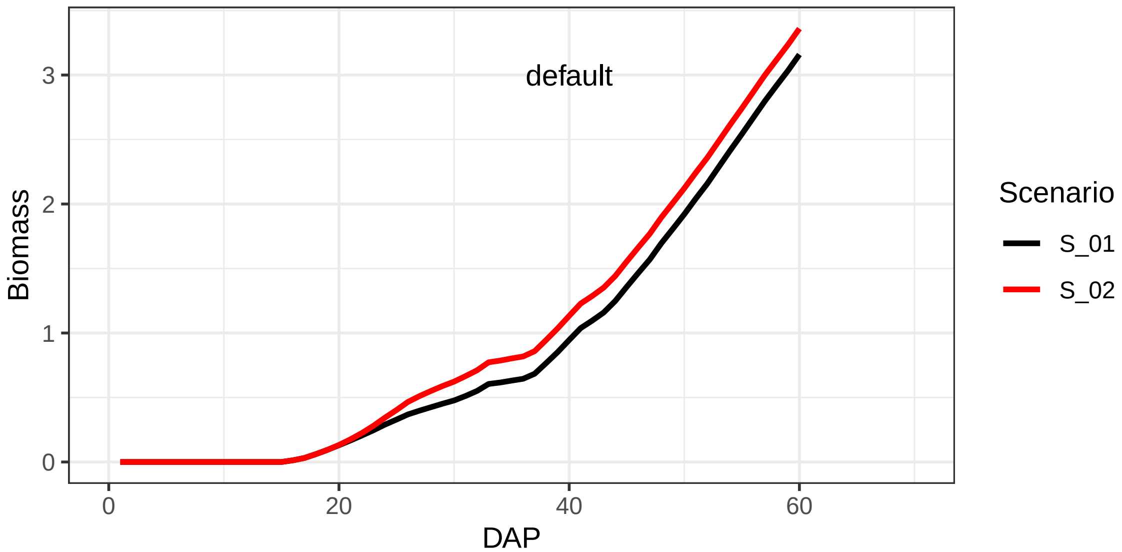
Finally, we can run AquaCrop using the `aquacrop_wrapper()` function and display a plot. The `param_values` argument is used to modify crop parameters from the default, which is the list provided as input to the `defaultpar` argument inside the `model_options` list. `situation` takes a vector of characters, that correspond to the `Scenario` variable in the `Scenario_s` tibble. The `daily_output` argument lets you choose which output type to return from the simulation (see Codes in AquaCrop manual p21-25). The `output` argument in the `model_options` defines the output format of the simulation. If `"croptimizr"`, the result is a list, ready to use in the CroptimizR package. If anything else, the result is a tibble.

Let us first run the simulation with the default parameters for spinach:

```r
default <- aquacrop_wrapper(param_values = list(),
               situation = c("S_01","S_02"),
               model_options = list(AQ = AQ, cycle_length = 70,
                                    defaultpar = Spinach, output = "df",
                                    daily_output = c(1,2)))

ggplot(mapping = aes(x=DAP, y=Biomass)) +
  geom_line(data = default,
            aes(group = Scenario, color = Scenario),
            linewidth = 1) +
```

```
  geom_text(aes(x=40, y=3, label = 'default'), color = "black") +
  xlim(0, 70) +
  scale_color_manual(values = c("black", "red")) +
  theme_bw()
```

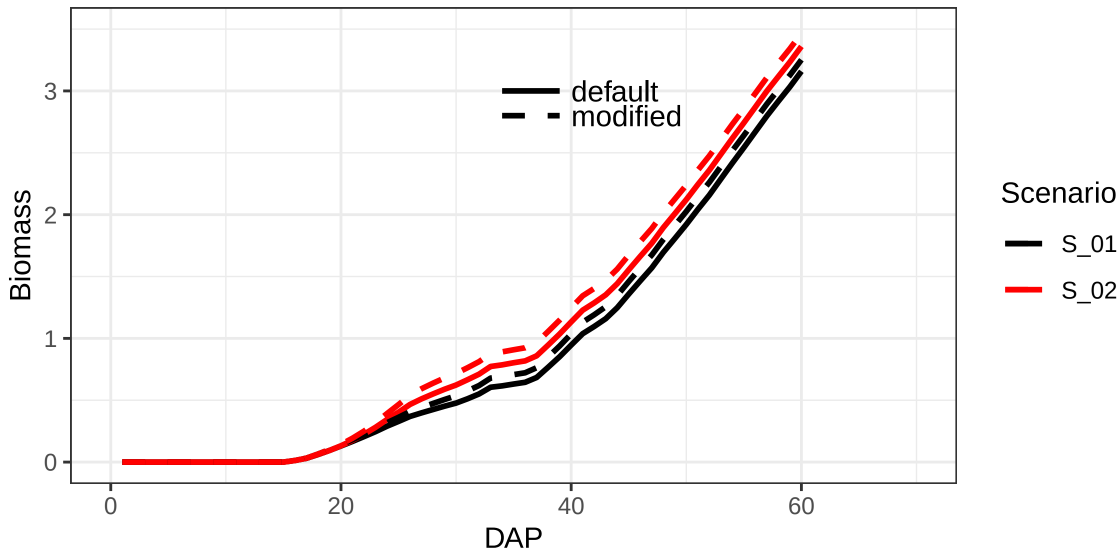


And now let's see what happens when we increase the canopy growth coefficient `cgc` from 0.15 to 0.18:

```
modified <- aquacrop_wrapper(param_values = list(cgc = 0.18),
               situation = c("S_01", "S_02"),
               model_options = list(AQ = AQ, cycle_length = 70,
                                    defaultpar = Spinach, output = "df",
                                    daily_output = c(1,2)))

ggplot() +
  ylim(0, 3.5) +
  geom_line(data = default,
            aes(x=DAP, y=Biomass, group = Scenario, color = Scenario),
            linewidth = 1) +
  geom_line(data = modified,
            aes(x=DAP, y=Biomass, group = Scenario, color = Scenario),
            linewidth = 1, linetype = 2) +
  geom_text(aes(x=40, y=3, label = 'default'),
            color = "black", hjust = 0) +
  geom_text(aes(x=40, y=2.8, label = 'modified'),
```

```r
             color = "black", hjust = 0) +
  geom_segment(mapping = aes(x = 34, xend = 39, y = 3, yend = 3),
                color = 'black', linewidth = 1, linetype = 1) +
  geom_segment(mapping = aes(x = 34, xend = 39, y = 2.8, yend = 2.8),
                color = 'black', linewidth = 1, linetype = 2) +
  xlim(0, 70) +
  scale_color_manual(values = c("black", "red")) +
  theme_bw()
```



A list of parameters with their explanation can be found in the `vignette("crop-parameters")`.